

Sales, Customer & Operations Analytics Deliverable

Domain Suite 2: Commercial Revenue, RFM Behavioral Segmentation & Manufacturing Quality

DELIVERED NET REVENUE

₹676.95 Cr

₹6,769,536,004 across 82,540 orders

AVERAGE ORDER VALUE

₹82,015

Delivered basket size

ACTIVE MASTER BUYERS

40,380

80.8% penetration of 50,000 base

CHAMPIONS SEGMENT

8,885

Highest recency & frequency buyers

STORE INVENTORY SLOTS

114,153

595 zero-stock OOS items

FACTORY SCRAP RATE

2.54%

640,075 rejected of 25.19M units

1. Commercial Sales Velocity & Merchandise Category Margins

Revenue contributions and realized profitability margins across major merchandise categories.

Category	Orders Count	Units Sold	Net Revenue (₹)	Gross Profit (₹)	Gross Margin (%)
Electronics	28,450	42,100	₹284.15 Cr	₹76.72 Cr	27.00%
Fashion & Apparel	22,100	58,400	₹148.90 Cr	₹52.12 Cr	35.00%
Home & Kitchen	16,800	34,200	₹115.40 Cr	₹34.62 Cr	30.00%
Grocery & Staples	9,850	89,500	₹82.50 Cr	₹14.85 Cr	18.00%
Health & Beauty	5,340	18,900	₹46.00 Cr	₹16.10 Cr	35.00%

2. Customer RFM Behavioral Segmentation & Loyalty Tiers

PostgreSQL RFM customer scoring distribution and loyalty tier spending velocity.

RFM Segment / Tier	Customer Count	Share (%)	Avg Recency	Total Spend (₹ Cr)	Strategic Retention Focus
Champions	8,885	17.77%	14 Days	₹234.80 Cr	VIP Exclusives & Early Access

RFM Segment / Tier	Customer Count	Share (%)	Avg Recency	Total Spend (₹ Cr)	Strategic Retention Focus
Loyal Customers	11,240	22.48%	28 Days	₹198.50 Cr	Cross-sell Category Expansion
Potential Loyalists	7,450	14.90%	45 Days	₹89.20 Cr	Second-Purchase Incentives
At Risk	6,518	13.04%	120 Days	₹68.40 Cr	Win-Back Personalized Discounts
Dormant / Lost	6,287	12.57%	240 Days	₹42.10 Cr	Reactivation Email Sequences

3. Operations: Inventory Health Matrix & Manufacturing Scrap Rates

Physical store stock health across 114,153 slots and defect rates across 10 manufacturing lines.

Operations Metric	Volume / Units	Share / Rate (%)	Operational Impact	Action
Adequate & Surplus Stock	96,678 Slots	84.69%	Healthy store availability	Maintain stock levels
Low Stock (At Reorder)	16,880 Slots	14.79%	Replenishment trigger reached	Generate DC transfer
Out of Stock (OOS)	595 Slots	0.52%	Immediate revenue loss risk	Priority expedited restock
Factory Work Orders	15,000 Batches	100.00%	25.19M finished units	Scheduled line balancing
Rejected Defect Units	640,075 Units	2.54% Scrap Rate	Within tolerance (< 3.0%)	Calibrate Line #4 & #8

Authoritative Production SQL Queries

Parameterized PostgreSQL execution queries powering all KPIs and exploratory analyses.

Business Purpose: Computes top-line net sales, order count, AOV, gross total, and absorbed promotional discounts.

```
SELECT
  COUNT(order_id) AS delivered_orders,
  ROUND(SUM(net_total), 2) AS total_net_revenue,
  ROUND(SUM(net_total) / 10000000.0, 2) AS revenue_crores,
  ROUND(AVG(net_total), 2) AS aov,
  ROUND(SUM(gross_total), 2) AS total_gross_revenue,
  ROUND(SUM(discount_amount), 2) AS total_discounts
FROM sales.orders
WHERE order_status = 'Delivered';
```

Business Purpose: Analyzes revenue, cost of goods sold (COGS), gross profit, and margin percentages across categories.

```
SELECT
  c.category_name,
  COUNT(DISTINCT oi.order_id) AS order_count,
  SUM(oi.quantity) AS total_units_sold,
  ROUND(SUM(oi.net_amount), 2) AS net_revenue,
  ROUND(SUM(oi.quantity * p.cost_price), 2) AS cogs,
  ROUND(SUM(oi.net_amount) - SUM(oi.quantity * p.cost_price), 2) AS gross_profit,
  ROUND((SUM(oi.net_amount) - SUM(oi.quantity * p.cost_price)) / NULLIF(SUM(oi.net_amount), 0) * 100, 2) AS
gross_margin_pct
FROM sales.order_items oi
JOIN sales.orders o ON oi.order_id = o.order_id
JOIN products.products p ON oi.prod_id = p.product_id
JOIN core.dim_brand b ON p.brand_id = b.brand_id
JOIN core.dim_category c ON b.category_id = c.category_id
WHERE o.order_status = 'Delivered'
GROUP BY c.category_name
ORDER BY net_revenue DESC;
```

Business Purpose: Profiles customer distribution, average recency, order frequency, and monetary spend across RFM segments.

```
SELECT
    rfm_segment,
    COUNT(*) AS customer_count,
    ROUND(AVG(recency_days), 1) AS avg_recency_days,
    ROUND(AVG(order_frequency), 1) AS avg_order_frequency,
    ROUND(SUM(monetary_value), 2) AS total_spend,
    ROUND(SUM(monetary_value) / 10000000.0, 2) AS spend_crores,
    ROUND(COUNT(*)::numeric / SUM(COUNT(*)) OVER () * 100, 2) AS customer_share_pct
FROM analytics.vw_customer_rfm_scores
GROUP BY rfm_segment
ORDER BY total_spend DESC;
```

Business Purpose: Identifies defect scrap rates and units rejected across all factory production lines.

```
SELECT
    pl.line_name,
    pl.supervisor_name,
    COUNT(wo.work_order_id) AS batches_completed,
    SUM(wo.quantity_produced) AS total_units_produced,
    SUM(wo.rejected_quantity) AS total_units_rejected,
    ROUND(SUM(wo.rejected_quantity)::numeric / NULLIF(SUM(wo.quantity_produced), 0) * 100, 2) AS
scrap_rate_pct
FROM manufacture.work_orders wo
JOIN manufacture.production_lines pl ON wo.line_id = pl.line_id
WHERE wo.status = 'Completed'
GROUP BY pl.line_name, pl.supervisor_name
ORDER BY scrap_rate_pct DESC;
```